

Shell Sort:

This method sorts separate subfiles of the original file. These subfiles contain every k^{th} element of the original file. The value of k is called an increment.

For example, if k is 5, the subfile consisting of $x[0]$, $x[5]$, $x[10]$,... is first sorted.

After the first k files are sorted, a new smaller value of k is chosen, and the file is again partitioned into a new set of subfiles. Each of these larger subfiles is sorted and the process is repeated again with a smaller value of k . The last value in this sequence must be 1.

For example, If the original file is,

25 57 48 37 12 92 86 33

and the sequence (5, 3, 1) $\rightarrow$ [increments] is chosen, the following subfiles are sorted on each iteration.

First iteration: (increment = 5)

($x[0]$, $x[5]$) ($x[1]$, $x[6]$) ($x[2]$, $x[7]$) ($x[3]$) ($x[4]$)

Second iteration: (increment = 3)

($x[0]$, $x[3]$, $x[6]$) ($x[1]$, $x[4]$, $x[7]$) ($x[2]$, $x[5]$)

Third iteration: (increment = 1)

($x[0]$, $x[1]$, $x[2]$, $x[3]$, $x[4]$, $x[5]$, $x[6]$, $x[7]$)

Illustration of shell sort as follows:

Each of the subfiles is sorted using simple insertion sort method.

Original file: 25, 57, 48, 37, 12, 92, 86, 33

I Pass (Span = 5) Or (increments =5)	25, 57, 48, 37, 12, 92, 86, 33
II Pass (Span = 3) Or (increments =3)	25, 57, 33, 37, 12, 92, 86, 48
III Pass (Span = 3) Or (increments =1)	25, 12, 33, 37, 48, 92, 86, 57
Sorted List	12, 25, 33, 37, 48, 57, 86, 92

```

void shell_sort(int a[], int n, int increments[], int numinc)
{
    int l,j,k,span,temp;
    for(k=0;k<numinc;k++)
    {
        span=increments[k];
        for(i=span;i<n;i++)
    
```

```
{
    temp=a[i];
    for(j=i-span; j>=0 && temp<a[j]; j=j-span)
    {
        a[j+span]=a[j];
    }
    a[j+span]=temp;
}
}
```